

AMAZON DATA ANALYSIS

Introduction

Amazon is a popular e-commerce shopping platform that allows customers to search, purchase, and receive products online. The application manages millions of customers, products, sellers, orders, payments, and deliveries.

To manage this large amount of information efficiently, Amazon requires a well-structured database management system.

This requirement analysis identifies the major data, entities, relationships, and operations required for an e-commerce database system based on the Amazon shopping application.

Understanding the E-Commerce Business Process

The Amazon shopping process can be explained as follows:

Step 1 – Customer Registration/Login

The customer creates an Amazon account or logs into an existing account using their email/mobile number and password.

Step 2 – Product Search

The customer searches for products using keywords, categories, filters, and sorting options.

Step 3 – Product Selection

The customer views product information such as:

- Product name
- Price
- Images
- Description
- Customer ratings
- Reviews
- Availability

Step 4 – Add to Cart

The customer selects the required product and adds it to the shopping cart.

Step 5 – Place Order

The customer reviews the cart, selects the delivery address, and confirms the order.

Step 6 – Payment

The customer selects an available payment method and completes the payment.

Step 7 – Order Processing

The seller receives the order and prepares the product for shipment.

Step 8 – Shipping and Delivery

The product is shipped and delivered to the customer's selected address.

Step 9 – Order Tracking

The customer can track the order and view its current status.

Step 10 – Review and Return

After receiving the product, the customer can provide a rating/review. If eligible, the customer can also request a return or refund.

Data Required for Customer Management

For an Amazon-like e-commerce application, the following customer information needs to be maintained.

Customer Data	Description
Customer_ID	Unique ID for each customer
Customer_Name	Name of the customer
Email	Registered email address
Phone_Number	Customer contact number
Password	Account authentication information
Address_ID	Unique ID for address
Delivery_Address	Address used for delivery
City	Customer's city
State	Customer's state
Pincode	Postal code

Customer Data Description

Registration_Date Date when account was created

Example:

Customer_ID : C101

Customer_Name : Rahul

Email : rahul@gmail.com

Phone : 9876543210

Address : 25, Main Road

City : Chennai

State : Tamil Nadu

Pincode : 600001

Major Entities

The major entities identified from the Amazon shopping process are:

Customer

The Customer entity stores information about people who purchase products from Amazon.

Attributes:

- **Customer_ID – Primary Key**
- **Customer_Name**
- **Email**
- **Phone_Number**
- **Password**
- **Registration_Date**

Product

The Product entity contains information about products available for purchase.

Attributes:

- **Product_ID – Primary Key**
- **Product_Name**

- **Description**
- **Price**
- **Category**
- **Stock_Quantity**
- **Seller_ID – Foreign Key**

Example: Mobile phones, laptops, books, clothing, etc.

Seller

The Seller entity contains information about sellers who list products on the platform.

Attributes:

- **Seller_ID – Primary Key**
- **Seller_Name**
- **Email**
- **Phone_Number**
- **Address**

A seller can sell multiple products.

Order

The Order entity stores information about products purchased by customers.

Attributes:

- **Order_ID – Primary Key**
- **Customer_ID – Foreign Key**
- **Order_Date**
- **Total_Amount**
- **Order_Status**
- **Delivery_Address**

Examples of order status:

- **Ordered**

- Shipped
- Out for Delivery
- Delivered
- Cancelled
- Returned

Payment

The Payment entity stores information about payments associated with orders.

Attributes:

- Payment_ID – Primary Key
- Order_ID – Foreign Key
- Payment_Date
- Amount
- Payment_Method
- Payment_Status

Payment methods may include online payment methods, cards, net banking, or cash on delivery where available.

Order Item

An order may contain multiple products. Therefore, an Order_Item entity can be used to connect orders and products.

Attributes:

- Order_ID – Foreign Key
- Product_ID – Foreign Key
- Quantity
- Price

Relationship Between Entities

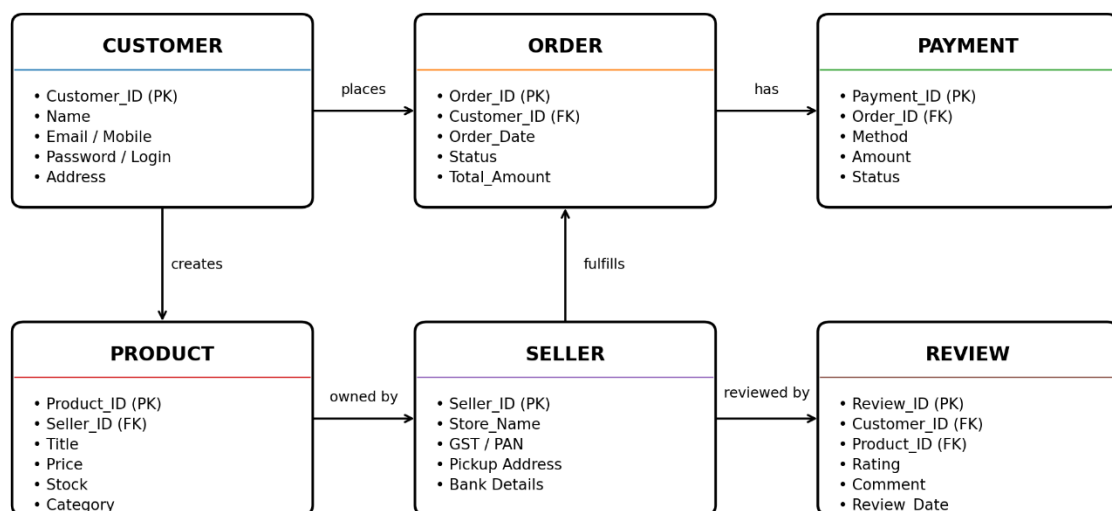
Relationship	Description
Customer → Order	A customer can place many orders
Order → Customer	Each order belongs to a customer
Seller → Product	A seller can sell many products
Product → Seller	A product is associated with a seller
Order → Product	An order can contain multiple products
Product → Order	A product can be included in multiple orders
Order → Payment	An order has payment information

Cardinality

- Customer 1 : N Order
- Seller 1 : N Product
- Order 1 : N Order_Item
- Product 1 : N Order_Item
- Order 1 : 1 Payment (*simplified model*)

Amazon E-Commerce Database — Conceptual ER Diagram

Requirement Analysis



Primary relationships: Customer → Order → Payment | Product → Seller | Customer → Review → Product
 PK = Primary Key FK = Foreign Key

PK = Primary Key | FK = Foreign Key

Major Entities: Customer, Order, Payment, Product, Seller, and Review.

Customer Data Operations

The Amazon-like application performs several operations on customer data.

1. Create

When a new customer registers, their details are added to the database.

Example:

New Customer → Customer Table

2. Read

The system retrieves customer information when the customer logs in, places an order, or views their account.

3. Update

Customers can update information such as:

- **Phone number**
- **Email**
- **Delivery address**
- **Account information**

4. Delete

Customer-related information may be removed or deactivated when an account is closed, subject to the platform's data-retention requirements.

5. Search

The system can identify a customer using information such as:

- **Customer ID**
- **Email**
- **Mobile number**

6. Authentication

The system verifies the customer's login credentials before providing access to their account.

Questions and Answers

1. Why does an e-commerce system require a database?

An e-commerce application like Amazon handles a very large amount of data every day. A database is required to store and manage:

- **Customer information**
- **Product information**
- **Seller details**
- **Orders**
- **Payments**
- **Delivery information**
- **Product stock**
- **Reviews and ratings**

The database helps Amazon retrieve information quickly, maintain data consistency, track orders, manage inventory, and process customer transactions efficiently.

2. What customer details need to be stored?

An Amazon-like e-commerce system needs to store:

- 1. Customer ID**
- 2. Customer name**
- 3. Email address**
- 4. Phone number**
- 5. Login/authentication information**
- 6. Delivery addresses**
- 7. Registration date**
- 8. Order history**

These details help the system identify customers, process orders, deliver products, and provide personalized account services.

3. What operations should be performed on customer data?

The major operations are:

Operation	Purpose
Create	Register a new customer
Read	Retrieve customer information
Update	Modify customer details
Delete/Deactivate	Remove or deactivate an account when appropriate
Search	Find customer information
Authentication	Verify customer login

These operations are commonly referred to as CRUD operations: Create, Read, Update, and Delete.

Conclusion

The analysis of the Amazon shopping application shows that an e-commerce system requires a strong database to manage its business operations.

The major entities identified are Customer, Product, Seller, Order, Payment, and Order_Item. These entities are connected through relationships that support the complete shopping process, from customer registration and product selection to payment, delivery, and order management.

Therefore, a properly designed e-commerce database improves data organization, transaction processing, inventory management, customer management, and overall system efficiency.